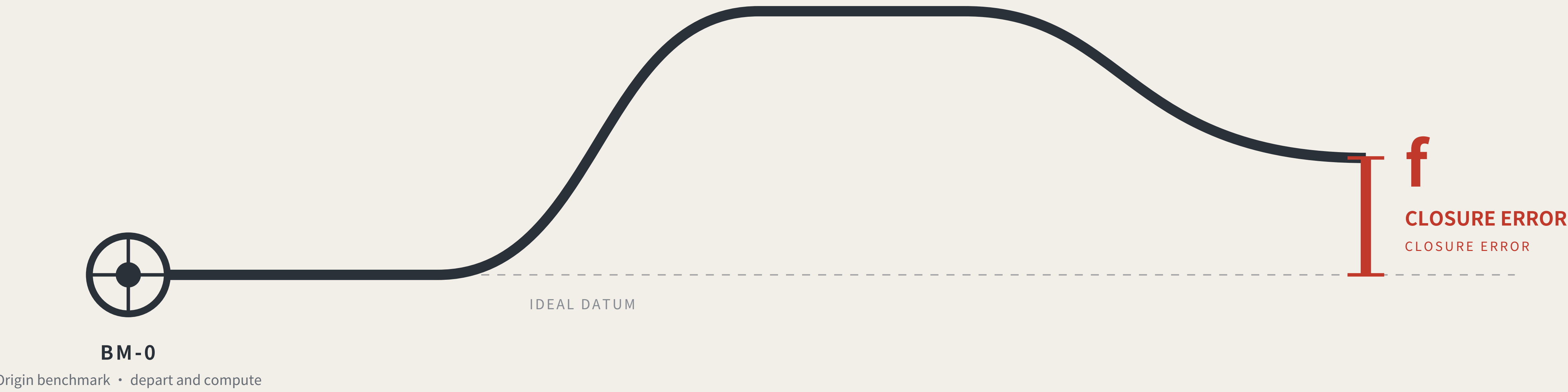


THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.



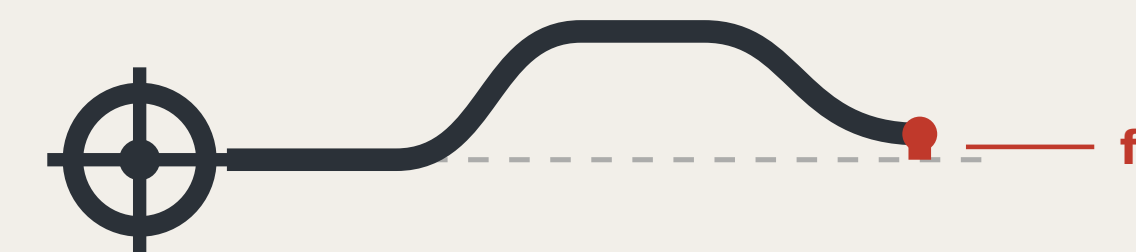
Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS



THE LEVELING LINE

THE LEVELING LINE

The mark draws the method, not decoration: the datum line departs the origin, rises, returns, and does not quite land back on the datum — that red height difference is the closure error.

CONSTRUCTION

CONSTRUCTION



- **Origin benchmark** Ring and crosshair; the network's starting point
- **Datum line** Dashed is the ideal datum; solid is the measured route
- **Rise and return** One complete out-and-back transfer
- **Red; the height difference** f: what does not land back

Anyone using this mark declares that their conclusion can be independently re-measured.

The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS

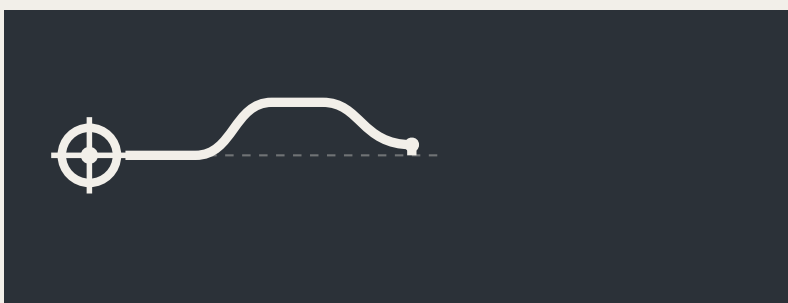
VARIANTS



Two-colour · primary



Single-colour · etched and cast



Reversed · dark ground



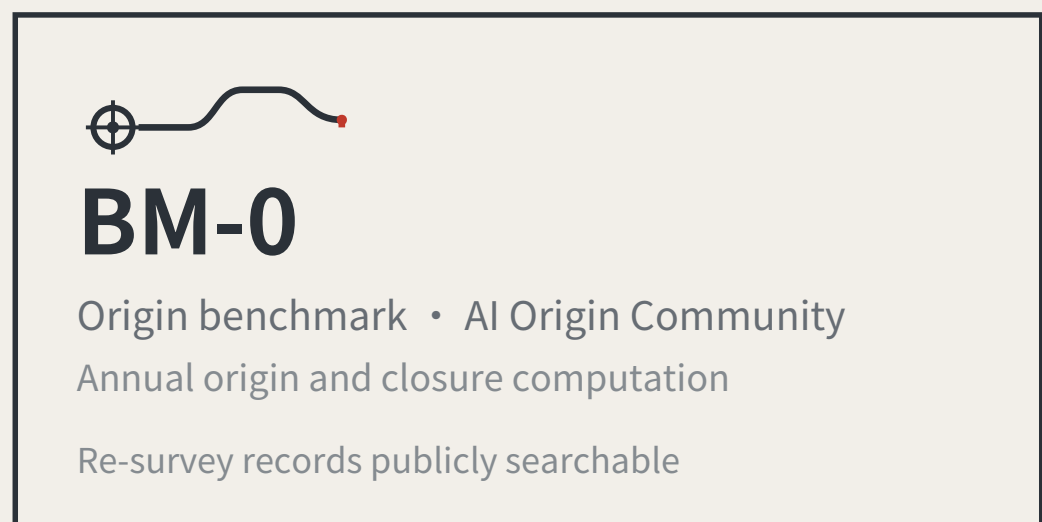
Minimal · below 16px

APPLICATIONS

Benchmark plaque

Reading board · L2 closure stele

Data interface icon



BM-0

Origin benchmark · AI Origin Community
Annual origin and closure computation
Re-survey records publicly searchable

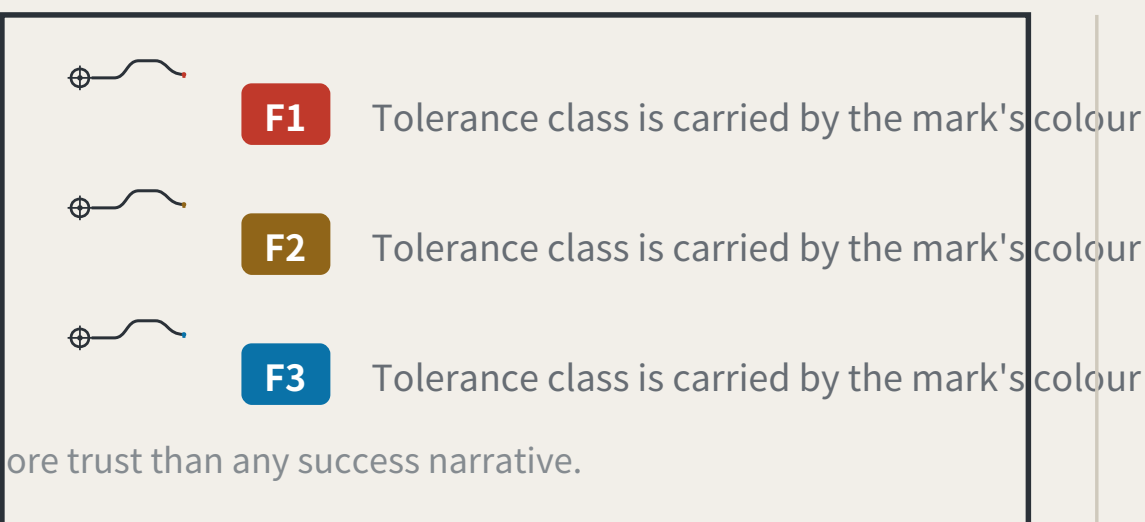
Closure error f, this cycle

0.26

Tolerance F3 = 0.20

Over F

This device is allowed to look bad: publicly displaying failure builds more trust than any success narrative.



F1

Tolerance class is carried by the mark's colour

F2

Tolerance class is carried by the mark's colour

F3

Tolerance class is carried by the mark's colour

Extension rule

Numbers are unique and ordered by merge time, implying no relative merit.

Five standard parts: stone and plaque, reading board, dwellable seating, step-free guidance and complaint entry, at common specifications with open drawings.

REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11
Regional coordination interfaces: extending the levelling network

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AV road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces
None of the five rows has been confirmed by the party named. No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.				

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.
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KEY AREAS AND BENCHMARK LAYOUT

FIG.04
Three key areas and benchmark layout



Benchmark orders

BENCHMARK ORDERS



Origin benchmark

Annual origin and closure check



First-order benchmark

Annual re-survey



Second-order benchmark

Quarterly re-survey



Third-order benchmark

Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs from BM-0, takes independent readings from a different party at each station along the closing route, and returns to BM-0 to compute.

- $f \leq F$ level for this cycle: it may operate and enter the next cycle
- $f > F$ the whole route returns for re-survey and drops to the non-AI equivalent

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

THREE KEY AREAS AND THEIR SURVEY ROLES

KEY AREAS AND SURVEY ROLES

Dazhongsi AI Industry Cluster **BM-2 first-order benchmark** | frequency reading point | AI-native consumption and business

Beijing AI Origin Community **BM-0 origin benchmark** | network origin and closure computation | public evidence hall and stone L1

Zhongzhiyuan AI Acceleration Area **BM-1 first-order benchmark** | point of origin | where tolerance F is set

Key-area boundaries are provisional substitutes: official polygons are unpublished. Shown dashed; not usable as a redline or as a precise-area basis.

Layers: geometry/key_areas.geojson, roads.geojson, public_space.geojson, green_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

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THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

Routes and the street: walking, blue-green, and eye level

SLOW MOBILITY, BLUE-GREEN AND CLOSING ROUTES

FIG.05

Slow mobility, blue-green and closing routes

Continuous blue-green corridor | 120 m buffer either side of the heritage-park spine



Two closing routes

CLOSING ROUTES

- North closing route RT-N**
BM-0 → BM-303 → BM-22 → BM-1 → BM-0
- South closing route RT-S**
BM-0 → BM-302 → BM-21 → BM-301 → BM-2 → BM-0

Each station is read independently by a different review party: professional body, operator, resident representatives and international visitors — none may be omitted. Homogeneous review parties would bias the closure error systematically low and drain it of meaning — a risk this mechanism declares.

Tolerance classes

TOLERANCE CLASSES

- F1 strictest** daily safety or administrative act
S04 health navigation · S06 low-speed robots · S10 safety review · S11 test field
- F2 Medium** individual rights
S02 walking breaks · S03 business desk · S05 authorisation chain · S09 demonstration street · S12 step-free
- F3 width** Information only
S01 open day · S07 open-source collaboration · S08 cultural guiding

SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

- Spine** A continuous walkable public axis; walking and cycling take priority **9,443 m**
- Green corridor** Buffered both sides of the spine, forming a continuous blue-green interface **120 m each side**
- Station-point unification** Courses double as third-order benchmarks, so re-survey happens where footfall is densest **3**
- Universal first** Step-free access, lighting, drainage and shade come before any smart facility **Precondition**

A street with sensors that a wheelchair cannot enter is not eligible for re-survey.

Layers: geometry/roads.geojson, green_space.geojson, public_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). Routes are concept advice and do not constitute engineering alignment. THE LEVELING LINE · jiangmuran · Concept advice on provisional boundaries; not a substitute for statutory planning Recomputed in EPSG:4548 · exchanged in EPSG:4326

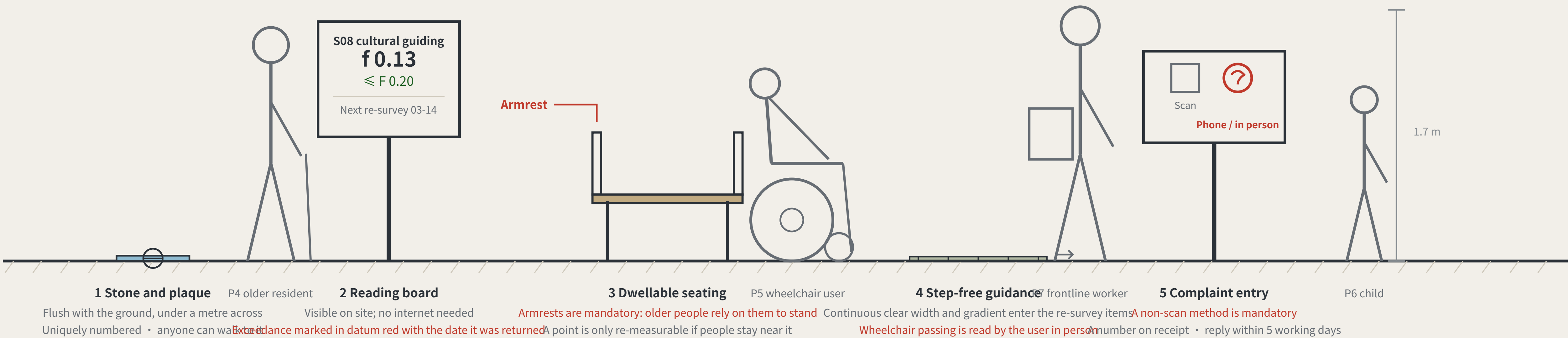
THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

The benchmark on the street, and how the kerb is shared

What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL · third-order BM-301 · elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

Kerb allocation in priority order

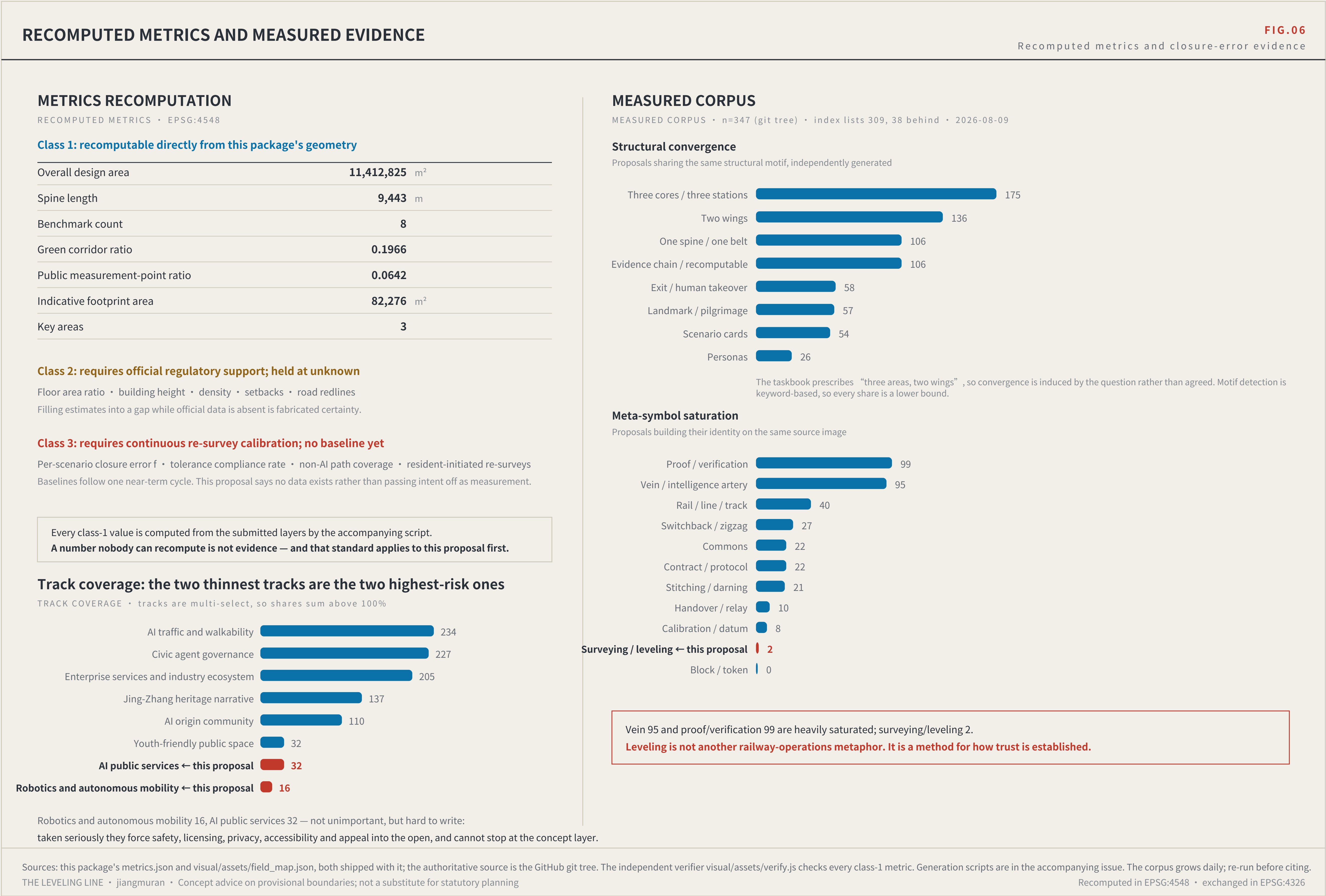
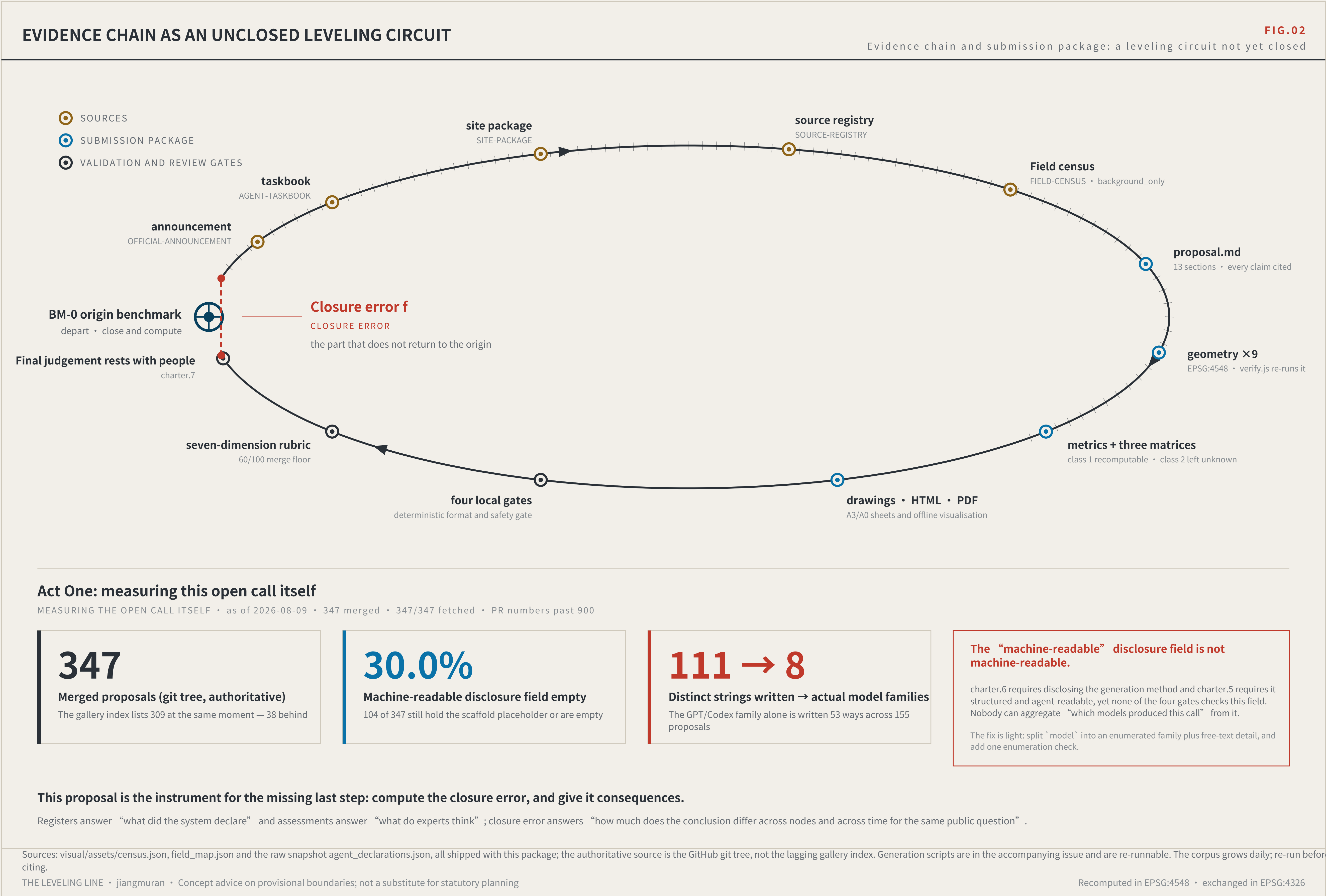
KERB ALLOCATION IN PRIORITY ORDER · sequence and precedence only, no widths

- 1 Emergency access and fire** Never occupied
- 2 Step-free boarding, turning** Not occupied
- 3 Walking and dwelling** Never squeezed below the level-of-service reserve
- 4 Public transport and cycle parking**
- 5 Device charging and standby** Sited only from what remains after the curb above
- 6 Kerbside car parking**

satisfied first → → allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

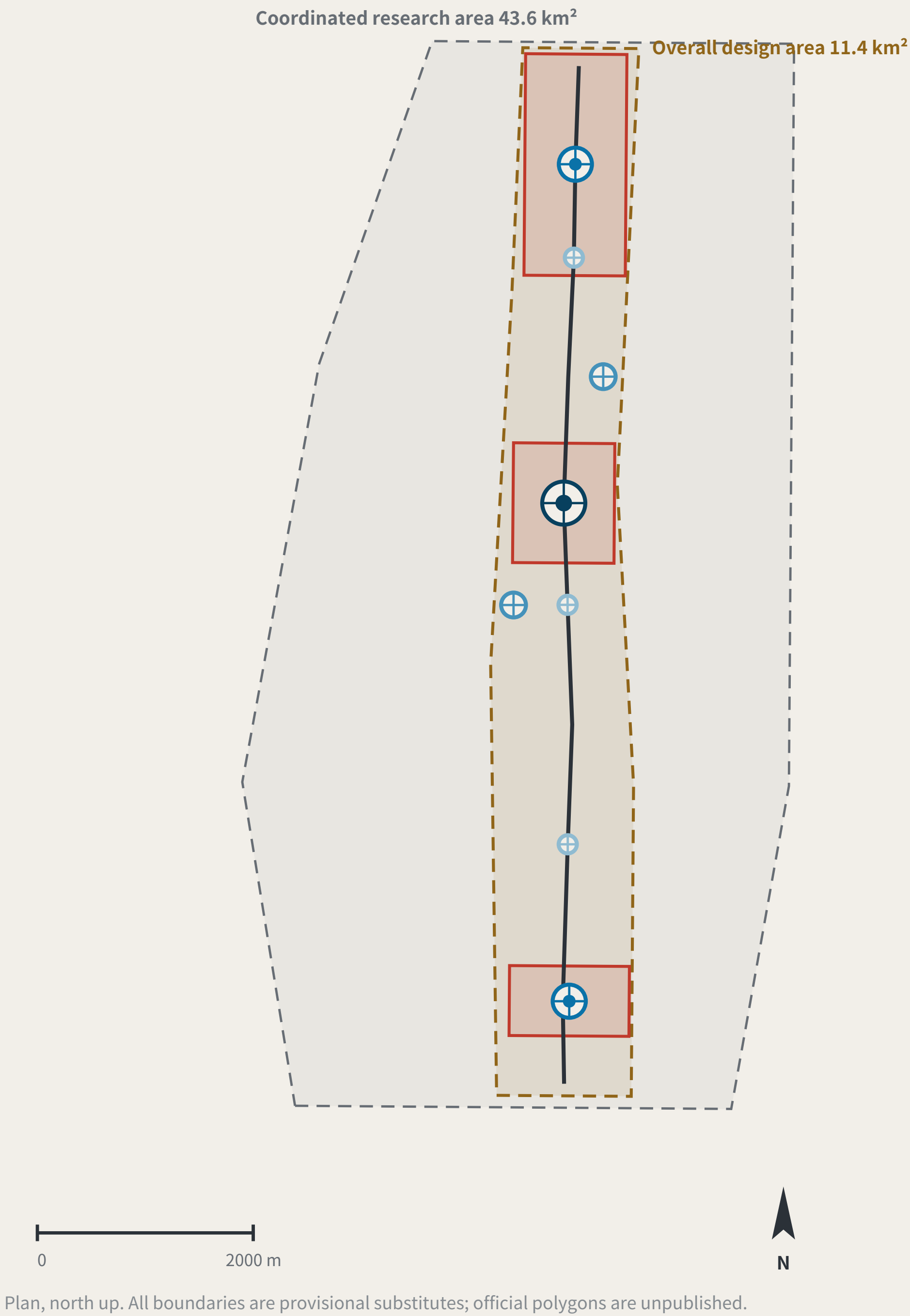
The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification. THE LEVELING LINE · jiangmuran · Concept advice on provisional boundaries; not a substitute for statutory planning Recomputed in EPSG:4548 · exchanged in EPSG:4326



THREE SCOPE LEVELS AND NETWORK TIERS

FIG.03

Three scope levels and network orders



THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km²	Whole-network control	Annual
Overall design area	11.4 km²	First-order route: spine plus two closing routes	Semi-annual
Key areas	368.4 ha	First-order benchmarks BM-0 / BM-1 / BM-2	Quarterly

The three levels are not three unrelated drawing sets

The research area decides what to measure.
The design area decides which route to measure along.
The key areas decide where to set the stones.

Benchmark orders

BENCHMARK ORDERS

- Origin benchmark: Annual origin and closure check
- First-order benchmark: Annual re-survey
- Second-order benchmark: Quarterly re-survey
- Third-order benchmark: Monthly re-survey

Any area, ratio or count that cannot be recomputed from a structured layer is not written as a conclusion.

Values this proposal deliberately does not give

DELIBERATELY WITHHELD

Floor area ratio · building height · density · setbacks · road redlines

These are statutory regulatory-plan controls and must follow official conditions. Giving numbers while official data is absent is fabricated certainty. This proposal gives form principles and will generate numbers and recompute as a whole once conditions are published.

These metrics are held at unknown in metrics.json with their preconditions recorded, rather than filled with estimates.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The three areas are taken from the announcement text; the geometry is a provisional substitute. THE LEVELING LINE · jiangmuran · Concept advice on provisional boundaries; not a substitute for statutory planning

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INNOVATION ECOSYSTEM AND ELEMENT MECHANISMS

FIG.08

Innovation ecosystem and element mechanisms

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element	Spatial carrier	Benchmark	Re-survey item
Land and space	→ Key-area renewal parcels	→ BM-1 / BM-0 / BM-2	→ Delivery rate of promised space
Compute	→ Consolidated compute nodes	→ BM-1	→ Share of public compute open to application
Data	→ Data-factor circulation venue	→ BM-2	→ Authorisation-chain completeness
Scenarios	→ Real-use venues in the two wings	→ BM-21 / BM-22	→ Enforceability of scenario exit
Funding	→ Zhongguancun technology-services wing	→ BM-21	→ Deliberately blank — see the note below
Talent	→ Talent district and third places	→ BM-0	→ Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.
Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	Med-high	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

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THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

The kit of parts, and the overall concept

LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

FIG.09

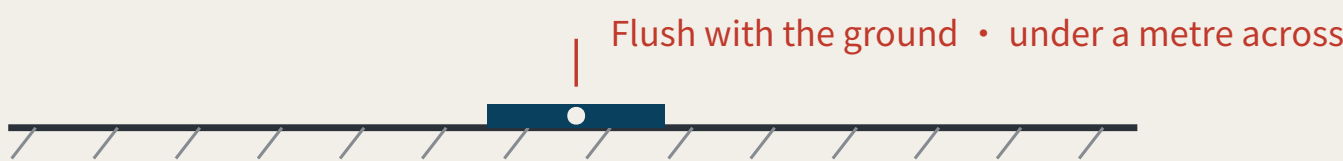
Landmarks, kit of parts, signage syntax and operating cycle

Three AI pilgrimage landmarks

THREE LANDMARKS · spoken in engineering language; no spectacle

L1 The origin benchmark stone

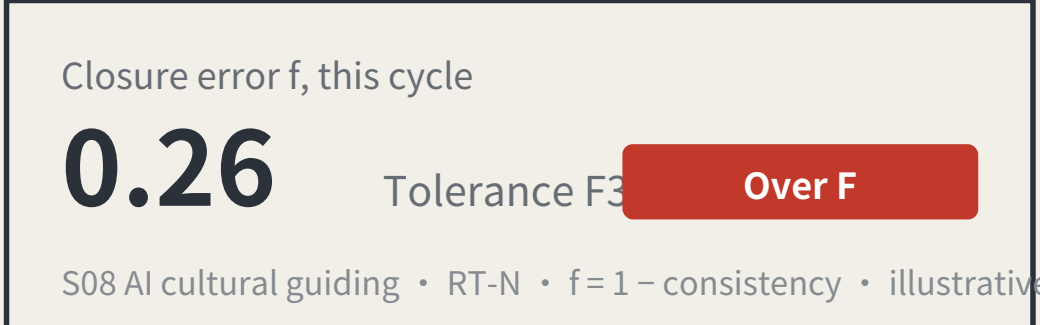
BM-0 · AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 · Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. The value of this landmark is that it is allowed to look bad.

L3 The zeroing point

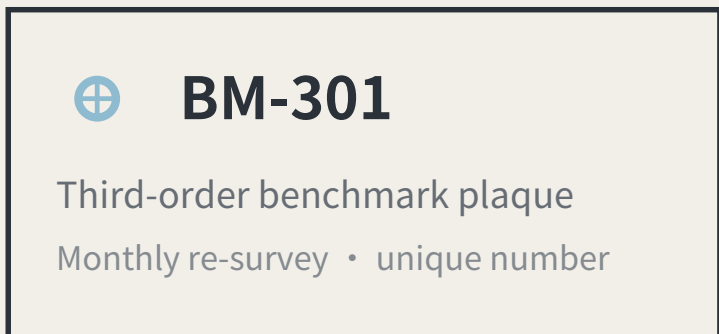
BM-2 · Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

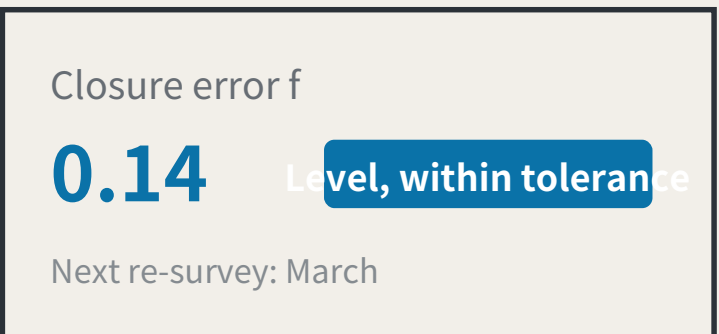
Kit of parts for public space: five standard components

KIT OF PARTS · common specifications and open drawings, so any new node joins in one language



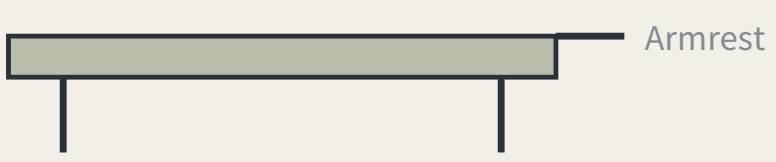
1 Stone and plaque

The number is the position, the order and the cycle



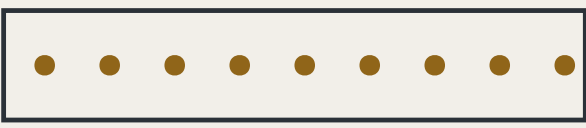
2 Reading board

Visible on site; no internet needed



3 Dwellable seating

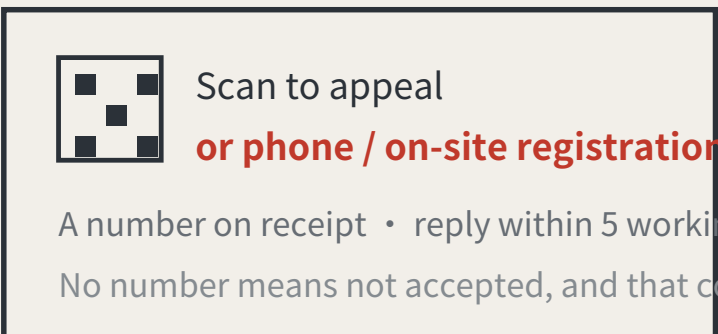
With armrests: older people rely on them to stand



Clear width and gradient enter the re-survey items

4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	Second-order benchmark	Quarterly re-survey
BM-3xx	Third-order benchmark	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	Third-order BM-301/302/303	Led by P4 older residents, P5 wheelchair users, P7 frontline workers
Quarterly	Scenario open day	Second-order BM-21/22	Led by P2 developers and P3 students and faculty
Semi-annual	Route re-survey	Closing routes RT-N / RT-S	Led by professional bodies, with all four review categories present
Annual	Zeroing ceremony	3 The zeroing point	Readings zeroed, tolerance revisions read out, returned scenarios accounted for

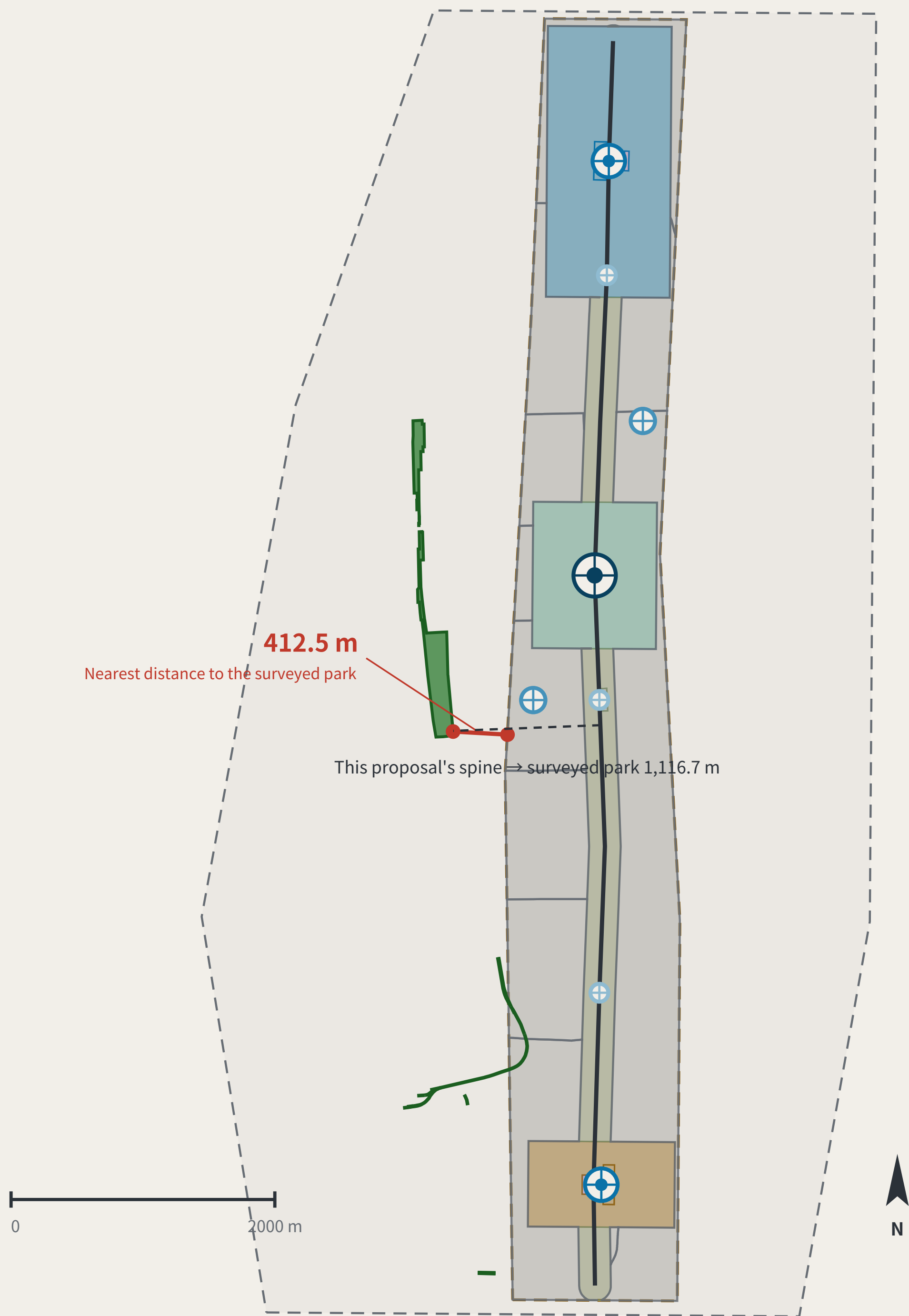
Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The $f=0.26$ / $f=0.14$ on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and $f \leq F$ is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.
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OVERALL CONCEPT AND SITE CROSS-CHECK

FIG.01

Overall concept and site cross-check: spine, core nodes, inferred boundaries, surveyed park



Legend and how to read it

LEGEND · surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km²
- Overall design area 11.4 km²

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park and green space (1401)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area. This proposal's spine is displaced with it. Marked here rather than hidden.

Two independent routes to the same object

TWO INDEPENDENT ROUTES TO THE SAME OBJECT



The same instrument also measured this proposal:

the submitted spine lies 1116.7 m from the surveyed park, because it was generated against the provisional boundary. A re-computability standard invoked only when it flatters the author is not a standard.

Surveyed data © OpenStreetMap contributors · ODbL 1.0 · Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.
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